

```
In [30]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

```
In [4]: df=pd.read_csv("abc.csv")
```

```
In [5]: df
```

```
Out[5]:
```

|     | User ID  | Gender | Age | EstimatedSalary | Purchased |
|-----|----------|--------|-----|-----------------|-----------|
| 0   | 15624510 | Male   | 19  | 19000           | 0         |
| 1   | 15810944 | Male   | 35  | 20000           | 0         |
| 2   | 15668575 | Female | 26  | 43000           | 0         |
| 3   | 15603246 | Female | 27  | 57000           | 0         |
| 4   | 15804002 | Male   | 19  | 76000           | 0         |
| ... | ...      | ...    | ... | ...             | ...       |
| 395 | 15691863 | Female | 46  | 41000           | 1         |
| 396 | 15706071 | Male   | 51  | 23000           | 1         |
| 397 | 15654296 | Female | 50  | 20000           | 1         |
| 398 | 15755018 | Male   | 36  | 33000           | 0         |
| 399 | 15594041 | Female | 49  | 36000           | 1         |

400 rows × 5 columns

```
In [9]: x=df.drop('User ID',axis=1)
```

```
In [11]: x=x.drop('Purchased',axis=1)
```

```
In [12]: x
```

Out[12]:

|            | Gender | Age | EstimatedSalary |
|------------|--------|-----|-----------------|
| <b>0</b>   | Male   | 19  | 19000           |
| <b>1</b>   | Male   | 35  | 20000           |
| <b>2</b>   | Female | 26  | 43000           |
| <b>3</b>   | Female | 27  | 57000           |
| <b>4</b>   | Male   | 19  | 76000           |
| ...        | ...    | ... | ...             |
| <b>395</b> | Female | 46  | 41000           |
| <b>396</b> | Male   | 51  | 23000           |
| <b>397</b> | Female | 50  | 20000           |
| <b>398</b> | Male   | 36  | 33000           |
| <b>399</b> | Female | 49  | 36000           |

400 rows × 3 columns

In [13]: `y=df['Purchased']`In [14]: `df.isnull().sum()`

Out[14]:

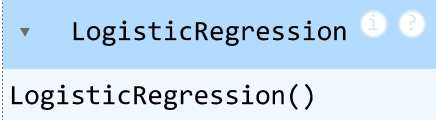
|                 |       |
|-----------------|-------|
| User ID         | 0     |
| Gender          | 0     |
| Age             | 0     |
| EstimatedSalary | 0     |
| Purchased       | 0     |
| dtype:          | int64 |

In [16]: `X=df[['Age','EstimatedSalary']]`  
`y=df['Purchased']`

In [17]: `from sklearn.model_selection import train_test_split`  
`X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.25,random_state=0)`

In [18]: `from sklearn.preprocessing import StandardScaler`  
`sc= StandardScaler()`  
`X_train=sc.fit_transform(X_train)`  
`X_test=sc.transform(X_test)`

In [19]: `from sklearn.linear_model import LogisticRegression`  
`logreg = LogisticRegression()`  
`logreg.fit(X_train,y_train)`

Out[19]:   
LogisticRegression()

```
In [20]: y_pred=logreg.predict(X_test)
```

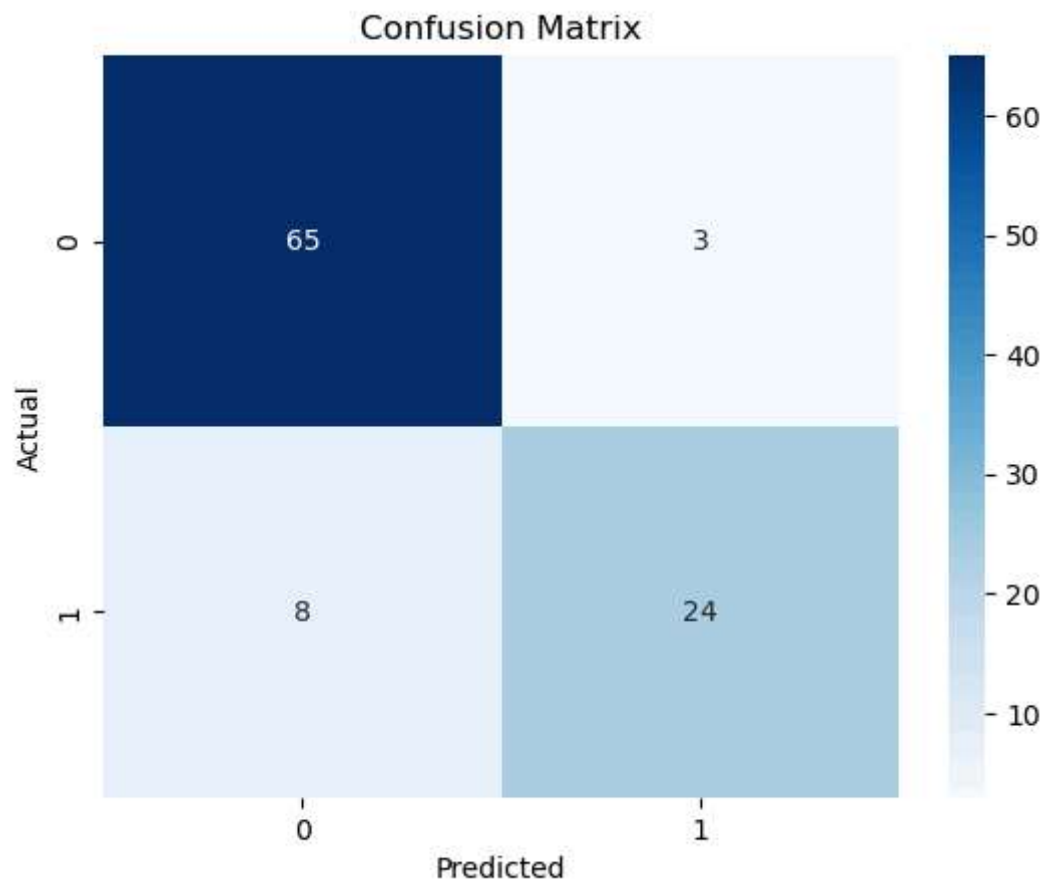
```
In [21]: from sklearn.metrics import confusion_matrix  
cm = confusion_matrix(y_test,y_pred)  
print(cm)
```

```
[[65  3]  
 [ 8 24]]
```

```
In [24]: from sklearn.metrics import accuracy_score,precision_score,recall_score  
accuracy = accuracy_score(y_test,y_pred)  
precision=precision_score(y_test,y_pred)  
recall= recall_score(y_test,y_pred)  
  
print("Accuracy:",accuracy)  
print("Precision:",precision)  
print("Recall:",recall)
```

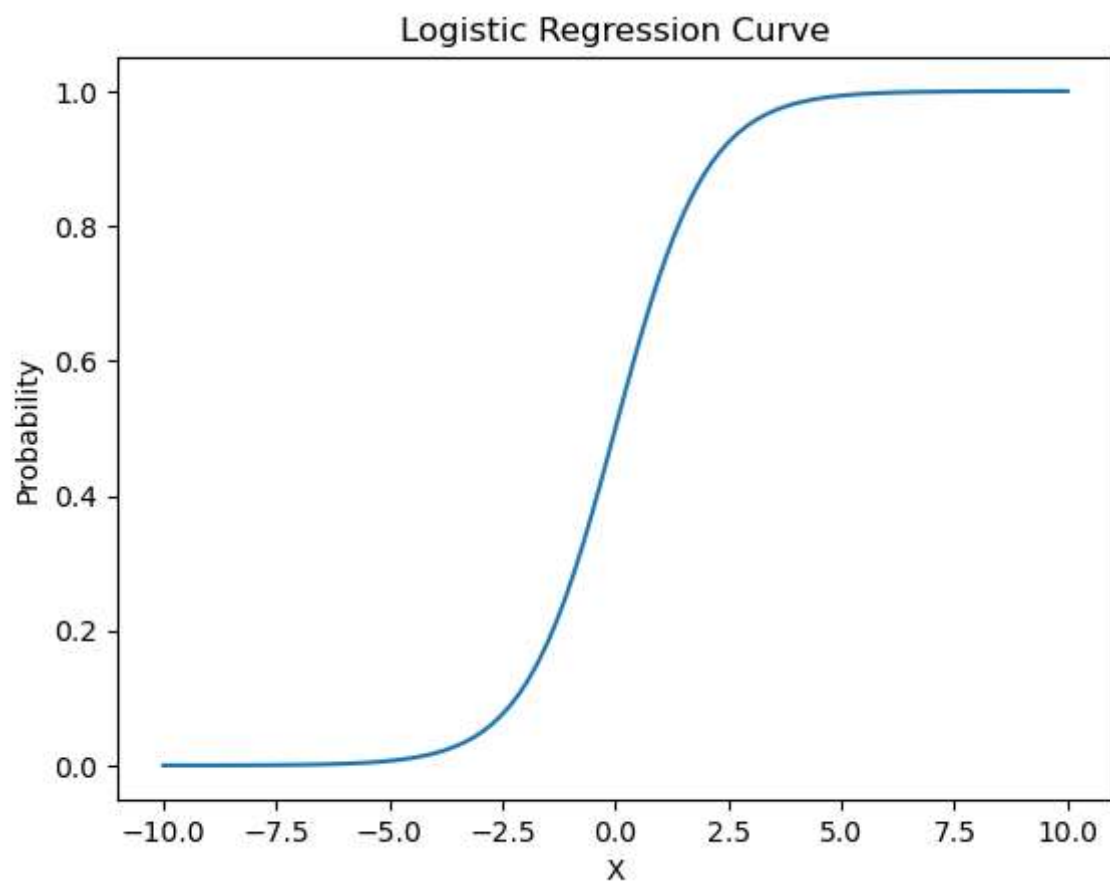
```
Accuracy: 0.89  
Precision: 0.8888888888888888  
Recall: 0.75
```

```
In [28]: sns.heatmap(cm,annot=True,fmt="d", cmap="Blues")  
plt.xlabel("Predicted")  
plt.ylabel("Actual")  
plt.title("Confusion Matrix")  
plt.show()
```



```
In [31]: x=np.linspace(-10,10,100)
         y=1/(1+np.exp(-x))
```

```
In [32]: plt.plot(x,y)
         plt.title("Logistic Regression Curve")
         plt.xlabel("X")
         plt.ylabel("Probability")
         plt.show()
```



In [ ]: